

# Mathematical Foundations of Political Methodology

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# Random Variables

- CDF
- Continuous Random Variables
- More on Expectation and Variance
- Brief introduction to inference

# Cumulative Distribution Function

## Definition

*The Cumulative Distribution Function (CDF) of a random variable  $X$  is:*

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

$$F_X(x) = \sum_{k \leq x} p_X(k)$$

The CDF characterizes how probability cumulates as  $X$  gets larger.

## Cumulative Distribution Function: Example

Suppose  $X \sim \text{Binomial}(n = 10, p = .3)$  What is the probability that  $x \leq 4$ ?

$$F_X(4) = P(X \leq 4)$$

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$F_X(4) = P(X \leq 4) = \sum_{x=0}^4 p_X(x)$$

$$F_X(4) = \sum_{k=0}^4 \binom{10}{k} .3^k (1 - .3)^{10-k} \approx .85$$

What is  $F_X(22)$ ?

What is  $F_X(-1)$ ?

# Cumulative Distribution Function

## Theorem

$F(x)$  is a CDF if and only if:

1  $F_X(x)$  is nondecreasing as  $x$  increases:

$$x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$$

2  $F_X(x)$  has limits at  $\pm\infty$

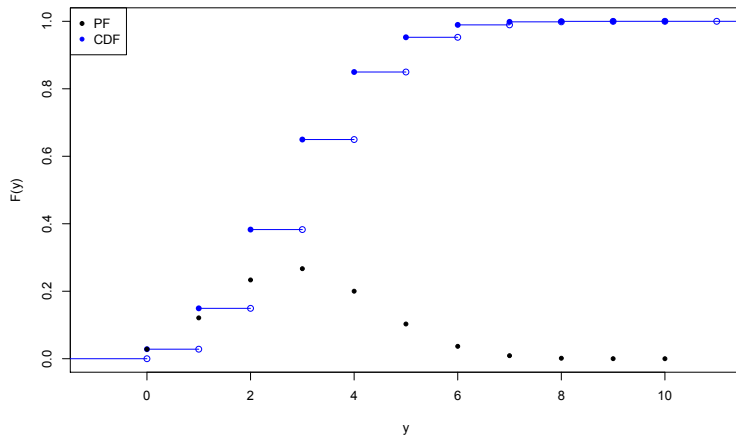
$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

3 Continuity from the right:

$$F_X(x) = \lim_{y \rightarrow x, y > x} F(y) \quad \forall x$$

# CDF



# Functions of Random Variables

We might (or often) apply a function to a random variable  $g(X)$ .  
How do we compute  $E[g(X)]$ ?

## Proposition

*Expected value of a function of a random variable: Suppose  $X$  is a discrete random variable that takes on values  $x_i$ ,  $i = \{1, 2, \dots\}$ , with probabilities  $p(x_i)$ . If  $g : X \rightarrow \mathcal{R}$ , then its expected value  $E[g(X)]$  is,*

$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

# Functions of Random Variables: Corollary

## Corollary

Suppose  $X$  is a random variable and  $a$  and  $b$  are **constants** (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

## Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= \sum_{x:p(x)>0} axp(x) + \sum_{x:p(x)>0} bp(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b(1) \end{aligned}$$



# Famous Distributions

- Bernoulli (coin flips)
- Binomial (sums of coin flips)
- Multinomial (multivariate coin flips)
- Poisson (sums of events over time or space)

Models of how world works.

# Poisson Distribution

## Definition

Suppose  $X$  is a random variable that takes on values  $X \in \{0, 1, 2, \dots\}$  and that  $P(X = k) = p(k)$  is,

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

for  $k \in \{0, 1, \dots\}$  and 0 otherwise. Then we will say that  $X$  follows a *Poisson* distribution with *rate* parameter  $\lambda$ .

$$X \sim \text{Poisson}(\lambda)$$

# Poisson Distribution

Properties:

1) It is a probability distribution.

■ Recall the **Taylor expansion** of  $e^x$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

■ Now expand the pdf of the poisson distribution:

$$\begin{aligned} e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} &= e^{-\lambda} \left( 1 + \lambda + \frac{\lambda^2}{2!} + \dots \right) \\ &= e^{-\lambda} (e^{\lambda}) = 1 \end{aligned}$$

# Poisson Distribution

- 2)  $E[X] = \lambda$ , that is, the parameter of the poisson distribution is its expectation

$$\begin{aligned} E[X] &= e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \end{aligned}$$

Define  $j = k - 1$ , then

$$\begin{aligned} E[X] &= e^{-\lambda} \lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= e^{-\lambda} \lambda e^{\lambda} \\ &= \lambda \end{aligned}$$

# Poisson Distribution

Properties:

3)  $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda} \lambda^k}{k!} \\ &= \lambda e^{-\lambda} \left( \sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{(k-1)!} \right) \end{aligned}$$

Let  $j = k - 1$ ,

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{(j+1) \lambda^j}{j!} \\ &= \lambda e^{-\lambda} \left( \sum_{j=0}^{\infty} \frac{(j) \lambda^j}{j!} + \sum_{j=0}^{\infty} \frac{(1) \lambda^j}{j!} \right) \\ &= \lambda e^{-\lambda} (\lambda e^{\lambda} + e^{\lambda}) \end{aligned}$$

# Poisson Distribution

## Properties

3)  $\text{var}(X) = \lambda$

$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

$$\text{var}(X) = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

# Joint PMFs

If we have more than one random variable, we can study their joint behavior.

If  $(x, y)$  are possible values of random variables  $X$  and  $Y$ , then

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

## Joint PMFs

If we have more than one random variable, we can study their joint behavior.

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## Joint PMFs

If we want to know the probability of a set of joint events  $(x, y) \in A$

$$P((X, Y) \in A) = \sum_{(x, y) \in A} p_{X,Y}(x, y)$$

We can also calculate the pmfs of  $X$  and  $Y$  individually (these are the marginal distributions:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

## Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy, Democracy}\}$$

$$X(\omega) = \begin{cases} 0 & \text{if } \omega = \text{Non-Democracy} \\ 1 & \text{if } \omega = \text{Democracy} \end{cases}$$

$$\Omega_2 = \{\text{not-Killed, Killed}\}$$

$$Y(\omega) = \begin{cases} y = 0 & \text{if } \omega = \text{not Killed} \\ y = 1 & \text{if } \omega = \text{Killed} \end{cases}$$

| $p_{X,Y}$ | 0     | 1     | $p_Y$ |
|-----------|-------|-------|-------|
| 1         | 0.034 | 0.005 | 0.039 |
| 0         | 0.466 | 0.495 | 0.961 |
| $p_X$     | 0.5   | 0.5   | 1     |

# Linearity

$$E[X + Y] = E[X] + E[Y]$$

Proof:

$$\begin{aligned} E[X + Y] &= \sum_x \sum_y (x + y) p(x, y) \\ &= \sum_x \sum_y x p(x, y) + \sum_x \sum_y y p(x, y) \\ &= \sum_x x \sum_y p(x, y) + \sum_y y \sum_x p(x, y) \\ &= \sum_x x p(x) + \sum_y y p(y) \end{aligned}$$

# Conditional Probability Distribution Function

## Definition

Suppose  $X$  and  $Y$  are discrete random variables with joint pdf  $p(x, y)$ . Then define the *conditional probability function*  $p(x|y)$  as

$$p(x|y) = \frac{p(x, y)}{p_Y(y)}$$

# Why Does Marginalization Work?

Begin with **discrete** case.

Consider jointly distributed discrete random variables,  $X$  and  $Y$ . We'll suppose they have joint pmf,

## A Table

|       | Y = 0              | Y = 1              |                    |
|-------|--------------------|--------------------|--------------------|
| X = 0 | p(0,0)             | p(0, 1)            | p <sub>X</sub> (0) |
| X = 1 | p(1,0)             | p(1,1)             | p <sub>X</sub> (1) |
|       | p <sub>Y</sub> (0) | p <sub>Y</sub> (1) |                    |
|       | Y = 0              | Y = 1              |                    |
| X = 0 | 0.01               | 0.05               | ?                  |
| X = 1 | 0.25               | 0.69               | ?                  |
|       | 0.26               | 0.74               |                    |

$$\begin{aligned}p_X(0) &= p(0|y = 0)p(y = 0) + p(0|y = 1)p(y = 1) \\&= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\&= 0.06\end{aligned}$$

$$\begin{aligned}p_X(1) &= p(1|y = 0)p(y = 0) + p(1|y = 1)p(y = 1) \\&= \frac{0.25}{0.26} \times 0.26 + \frac{0.69}{0.74} \times 0.74\end{aligned}$$

## Definition

*Two random variables  $X$  and  $Y$  are independent if for any two sets of real numbers  $A$  and  $B$ ,*

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$$

*Equivalently we will say  $X$  and  $Y$  are independent if,*

$$p(x, y) = p_X(x)p_Y(y)$$

*If  $X$  and  $Y$  are not independent, we will say they are **dependent***

# Conditional Distribution

If  $X$  and  $Y$  are independent, then

$$\begin{aligned} p_{X|Y}(x|y) &= \frac{p(x, y)}{p_Y(y)} \\ &= \frac{p_X(x)p_Y(y)}{p_Y(y)} \\ &= p_X(x) \end{aligned}$$

In words: the distribution of  $X$  does not change as levels of  $Y$  change.



# Conditional Expectation

$$\mathbb{E}(Y|X) = \sum_y y \cdot p_{Y|X}(y|x)$$

$\mathbb{E}(Y|X)$  is the *best* way to predict  $Y$  given  $X$ .

# Continuous Random Variables

Many times we deal with continuous events:

- Growth rate of an economy.
- Length of a recession.
- Where a ball goes through a soccer net.

We will need continuous random variables to assign numbers to these events.

# Probability of a continuous random variable

## Definition

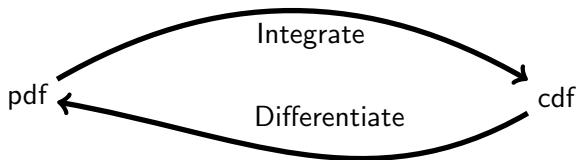
*If  $X$  is a continuous random variable, the probability density function of  $X$  is the function  $f_X(x)$  that satisfies.*

$$\begin{aligned}F_X(x) &= P(X \leq x) \\&= P(X \in (-\infty, x)) \\&= \int_{-\infty}^x f_X(t) dt \quad x \in \mathbf{R}\end{aligned}$$

$$\int_a^b f_X(x) dx = P(X \in (a, b))$$

# Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.



## RV: continuous uniform

### Definition

*We can define  $Y$  to have a uniform distribution on the interval  $(a, b)$*

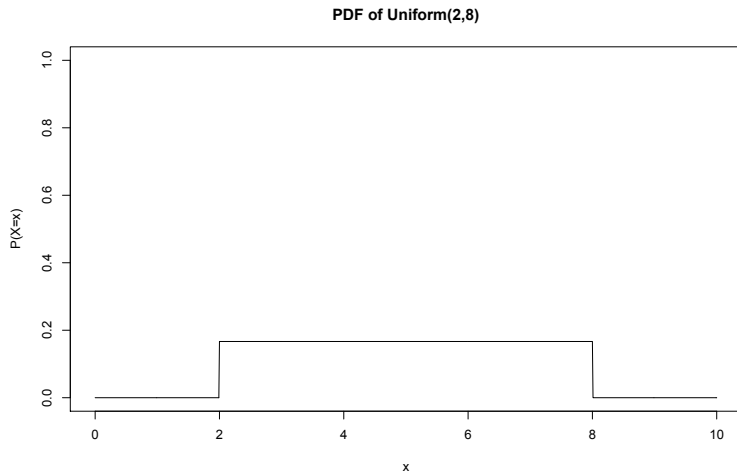
$$f_Y(y) = \begin{cases} \frac{1}{b-a} & \text{if } a < y < b \\ 0 & \text{otherwise} \end{cases}$$

## Example: Uniform

Example: Suppose that we are waiting for Comcast to show up and install our cable package. They say that they may arrive between 2:00 and 8:00. Without any further information, you may have no reason to suspect any particular time over any other.

$$X \sim U(2, 8)$$

## RV: continuous uniform

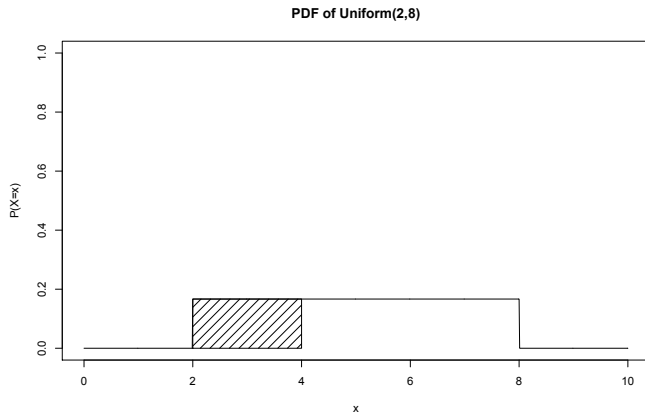


## RV: continuous uniform

What is the probability that the cable installation truck arrives before 4:00?



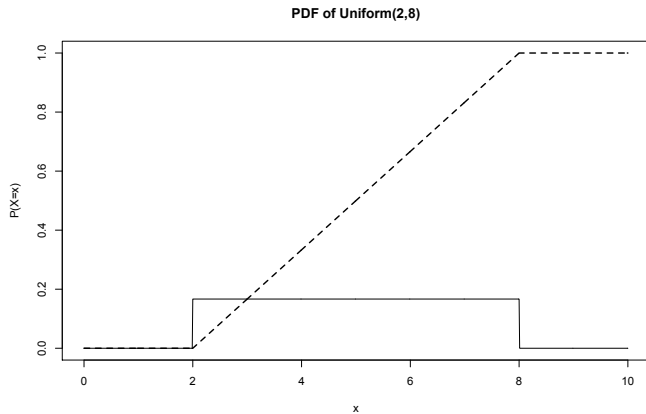
RV: continuous uniform ( $P(X \leq 4)$ )



## RV: continuous uniform

$$\begin{aligned}P(Y \leq y) &= \int_{x=-\infty}^y f(x) dx = \int_{x=-\infty}^y \frac{1}{b-a} dx = \int_{x=a}^y \frac{1}{b-a} dx \\&= \frac{y}{b-a} - \frac{a}{b-a} = \frac{y-a}{b-a} \\&= \frac{4-2}{8-2}\end{aligned}$$

## RV: continuous uniform



# Common Continuous Distributions: Exponential

Example: Suppose that the average durability of a Peace Accord in the Democratic Republic of Congo is 6 months. How can we model the time to failure of Peace Accord?

The exponential distribution is often used to model 'time to failure' processes.

$X \sim \exp(\lambda)$  if its PDF is:

$$f(x) = \lambda e^{-\lambda x}$$

with  $f(x) = 0$  for  $x < 0$

$\lambda$  is known as the 'rate' parameter.

$$\lambda = \frac{1}{E(X)}$$

# Exponential CDF

The CDF of  $X \sim \exp(\lambda)$  is:

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx = \int_0^x \lambda e^{-\lambda x} dx \\ &= F(x) - F(0) \end{aligned}$$

where  $F(x)$  is the antiderivative of  $f(x)$ .

$$\begin{aligned} &= -\frac{\lambda e^{-\lambda x}}{\lambda} - \left(-\frac{\lambda e^{-\lambda \cdot 0}}{\lambda}\right) \\ &= 1 - e^{-\lambda x} \end{aligned}$$

# Expectation With Continuous Random Variables

## Definition

*If  $X$  is a continuous random variable then,*

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

Suppose  $X \sim \text{Uniform}(0, 1)$ . What is  $E[X]$ ?

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{-\infty}^0 x0dx + \int_0^1 x1dx + \int_1^{\infty} x0dx \\ &= 0 + \frac{x^2}{2} \Big|_0^1 + 0 \\ &= 0 + \frac{1}{2} + 0 \\ &= \frac{1}{2} \end{aligned}$$

# Expectations of Functions

## Proposition

*Suppose  $X$  is a continuous random variable and  $g : \mathbb{R} \rightarrow \mathbb{R}$ . Then,*

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$



# Expectations of Functions

Suppose  $g(X) = X^2$  and  $X \sim \text{Uniform}(0, 1)$ . What is  $E[g(X)]$ ?

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x)f(x)dx \\ &= \int_0^1 x^2 \frac{1}{1-0} dx \\ &= \frac{x^3}{3} \Big|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

## Corollary

Suppose  $X$  is a continuous random variable. Then,

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b)f(x)dx \\ &= a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx \\ &= aE[X] + b \times 1 \end{aligned}$$



## Definition

*Variance. If  $X$  is a continuous random variable, define its variance,  $\text{Var}(X)$ ,*

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] \\ &= \int_{-\infty}^{\infty} (x - E[X])^2 f(x) dx \\ &= E[X^2] - E[X]^2\end{aligned}$$

## Variance: Random Variable

$X \sim \text{Uniform}(0, 1)$ . What is  $\text{Var}(X)$ ?

$$\begin{aligned} E[X^2] &= \frac{1}{3} \\ E[X]^2 &= \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

## Definition

Suppose  $X$  is a random variable with  $X \in \mathbf{R}$  and probability density function

$$f(x) = \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Then  $X$  is a **normally** distributed random variable with parameters  $\mu$  and  $\sigma^2$ .

Equivalently, we'll write

$$X \sim \text{Normal}(\mu, \sigma^2)$$

# Support for President Trump

Suppose we are interested in modeling **presidential approval**

- Let  $Y$  represent random variable: proportion of population who “approves job president is doing”
- Individual responses (that constitute proportion) are **independent** and **identically** distributed (sufficient, not necessary) and we take the average of those individual responses
- Observe **many** responses ( $N \rightarrow \infty$ )
- Then (by Central Limit Theorem)  $Y$  is **Normally** distributed, or

$$Y \sim \text{Normal}(\mu, \sigma^2)$$
$$f(y) = \frac{\exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

# Expected Value/Variance of Normal Distribution

$Z$  is a standard normal distribution if

$$F_Z(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-z^2/2) dz$$

## Proposition

*Scale/Location.* If  $Z \sim N(0, 1)$ , then  $X = aZ + b$  is,

$$X \sim \text{Normal}(b, a^2)$$

Proof:  $Z \sim N(0, 1)$  and  $Y = aZ + b$ , then  $Y \sim N(b, a^2)$

To prove we need to show that density for  $Y$  is a normal distribution.

That is, we'll show  $F_Y(x)$  is Normal cdf.

Call  $F_Z(x)$  cdf for standardized normal.

$$\begin{aligned} F_Y(x) &= P(Y \leq x) \\ &= P(aZ + b \leq x) \\ &= P\left(Z \leq \left[\frac{x - b}{a}\right]\right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-b}{a}} \exp\left(-\frac{z^2}{2}\right) dz \\ &= F_Z\left(\frac{x - b}{a}\right) \end{aligned}$$



Proof:  $Z \sim N(0, 1)$  and  $Y = aZ + b$ , then  $Y \sim N(b, a^2)$

So, we can work with  $F_Z(\frac{x-b}{a})$ .

$$\begin{aligned}\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\ &= f_Z(\frac{x-b}{a}) \frac{1}{a} \quad \text{By the chain rule} \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(\frac{x-b}{a})^2}{2}\right] \quad \text{By definition of } f_Z(x) \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(x-b)^2}{2a^2}\right] \\ &= \text{Normal}(b, a^2)\end{aligned}$$

# Expectation and Variance

Assume we know:

$$\begin{aligned}E[Z] &= 0 \\ \text{Var}(Z) &= 1\end{aligned}$$

This implies that, for  $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned}E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \\ \text{Var}(Y) &= \text{Var}(\sigma Z + \mu) \\ &= \sigma^2 \text{Var}(Z) + \text{Var}(\mu) \\ &= \sigma^2 + 0 \\ &= \sigma^2\end{aligned}$$

## Back To Trump

Suppose  $\mu = 0.39$  and  $\sigma^2 = 0.0025$

$P(Y \geq 0.45)$

$$\begin{aligned}P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\&= 1 - P(0.05Z + 0.39 \leq 0.45) \\&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05}) \\&= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{6/5} \exp(-z^2/2) dz \\&= 1 - F_Z\left(\frac{6}{5}\right) \\&= 0.1150697\end{aligned}$$

# The Gamma Function

## Definition

Suppose  $\alpha > 0$ . Then define  $\Gamma(\alpha)$  as

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

- For  $\alpha \in \{1, 2, 3, \dots\}$   
 $\Gamma(\alpha) = (\alpha - 1)!$
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

# Gamma Distribution

Suppose we have  $\Gamma(\alpha)$ ,

$$\frac{\Gamma(\alpha)}{\Gamma(\alpha)} = \frac{\int_0^\infty y^{\alpha-1} e^{-y} dy}{\Gamma(\alpha)}$$
$$1 = \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1} e^{-y} dy$$

Set  $X = Y/\beta$

$$\begin{aligned} F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\ &= P(Y \leq x\beta) \\ &= F_Y(x\beta) \\ \frac{\partial F_Y(x\beta)}{\partial x} &= f_Y(x\beta)\beta \end{aligned}$$

The result is:

$$f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}$$

## Definition

*Suppose  $X$  is a continuous random variable, with  $X \geq 0$ . Then if the pdf of  $X$  is*

$$f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}$$

*if  $x \geq 0$  and 0 otherwise, we will say  $X$  is a Gamma distribution.*

$$X \sim \text{Gamma}(\alpha, \beta)$$

## Gamma Distribution

Suppose  $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned} \tag{0.2}$$

Suppose  $\alpha = 1$  and  $\beta = \lambda$ . If

$$\begin{aligned} X &\sim \text{Gamma}(1, \lambda) \\ f(x|1, \lambda) &= \lambda e^{-x\lambda} \end{aligned}$$

We will say

$$X \sim \text{Exponential}(\lambda)$$

# Properties of Gamma Distributions

## Proposition

*Suppose we have a sequence of independent random variables, with*

$$X_i \sim \text{Gamma}(\alpha_i, \beta)$$

*Then*

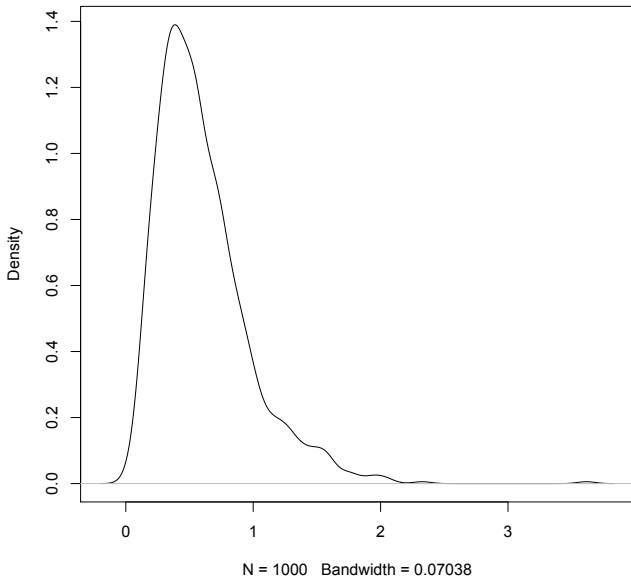
$$Y = \sum_{i=1}^N X_i$$

$$Y \sim \text{Gamma}(\sum_{i=1}^N \alpha_i, \beta)$$



We can evaluate in R with `dgamma` and simulate with `rgamma`  
 $X \sim \text{Gamma}(3, 5)$  and we evaluate at 3,  
`dgamma(3, shape= 3, rate = 5)`  
and we can simulate with  
`rgamma(1000, shape = 3, rate = 5)`

`density.default(x = rgamma(1000, shape = 3, rate = 5))`



## $\chi^2$ Distribution

Suppose  $Z \sim \text{Normal}(0, 1)$ .

Consider  $X = Z^2$

$$\begin{aligned}F_X(x) &= P(X \leq x) \\&= P(Z^2 \leq x) \\&= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\&= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz \\&= F_Z(\sqrt{x}) - F_Z(-\sqrt{x})\end{aligned}$$

$$\frac{\partial F_X(x)}{\partial x} = f_Z(\sqrt{x}) \frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x}) \frac{1}{2\sqrt{x}}$$

## $\chi^2$ Distribution

$$\begin{aligned}\frac{\partial F_X(x)}{\partial x} &= f_Z(\sqrt{x})\frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x})\frac{1}{2\sqrt{x}} \\&= \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{2\pi}}(2e^{-\frac{x}{2}}) \\&= \frac{1}{\sqrt{x}}\frac{1}{\sqrt{2\pi}}(e^{-\frac{x}{2}}) \\&= \frac{(\frac{1}{2})^{1/2}}{\Gamma(\frac{1}{2})} \left( x^{1/2-1} e^{-\frac{x}{2}} \right)\end{aligned}$$

$X \sim \text{Gamma}(1/2, 1/2)$

Then if  $X = \sum_{i=1}^N Z^2$

$X \sim \text{Gamma}(n/2, 1/2)$

## Definition

Suppose  $X$  is a continuous random variable with  $X \geq 0$ , with pdf

$$f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}$$

Then we will say  $X$  is a  $\chi^2$  distribution with  $n$  degrees of freedom. Equivalently,

$$X \sim \chi^2(n)$$

## $\chi^2$ Properties

Suppose  $X \sim \chi^2(n)$

$$E[X] = E \left[ \sum_{i=1}^N Z_i^2 \right]$$

$$= \sum_{i=1}^N E[Z_i^2]$$

$$\text{var}(Z_i) = E[Z_i^2] - E[Z_i]^2$$

$$1 = E[Z_i^2] - 0$$

$$E[X] = n$$

## $\chi^2$ Properties

$$\begin{aligned}\text{var}(X) &= \sum_{i=1}^N \text{var}(Z_i^2) \\ &= \sum_{i=1}^N (E[Z_i^4] - E[Z_i]^2) \\ &= \sum_{i=1}^N (3 - 1) = 2n\end{aligned}$$

We will use the  $\chi^2$  in all sorts of statistics courses

# Student's $t$ -Distribution

## Definition

Suppose  $Z \sim \text{Normal}(0, 1)$  and  $U \sim \chi^2(n)$ . Define the random variable  $Y$  as,

$$Y = \frac{Z}{\sqrt{\frac{U}{n}}}$$

If  $Z$  and  $U$  are independent then  $Y \sim t(n)$ , with pdf

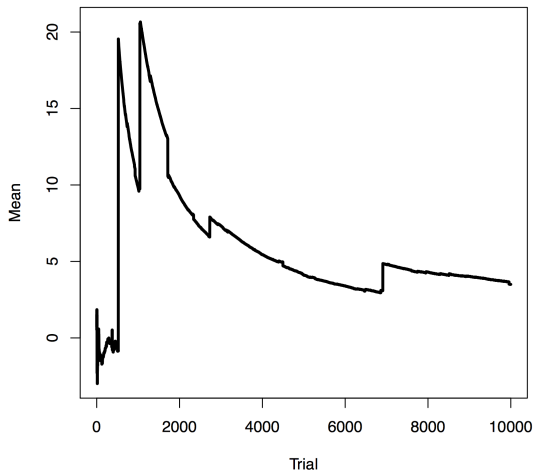
$$f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

We will use the  $t$ -distribution extensively for *test-statistics*



# Student's $t$ -Distribution, Properties

Suppose  $n = 1$ , **Cauchy** distribution



then:

If  $X \sim \text{Cauchy}(1)$ ,

# Student's $t$ -Distribution, Properties

Suppose  $n > 2$ , then

$$\text{var}(X) = \frac{n}{n-2}$$

As  $n \rightarrow \infty$   $\text{var}(X) \rightarrow 1$ .

# Using the t-Distribution

We often construct **test-statistics**

Suppose we take  $N$  iid draws,

$$X_i \sim \text{Normal}(\mu, \sigma^2)$$

Define our data set  $\mathbf{x} = (x_1, \dots, x_N)$

Calculate:

$$\bar{x} = \sum_{i=1}^N \frac{x_i}{N}$$

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

# Estimators

- Our data are realizations from a sample space.
- Functions of data are a kind of random variable called 'statistics'.
- If that function is used to infer the value of a parameter of a distribution, it is called an 'estimator'.
- All estimators are statistics, and so they are random variables.

$\bar{x}(X)$  is a function called the sample mean, and it produces estimates of the true mean ( $\mu$ )

$s^2(X)$  is a function called the sample variance, and it produces estimates of the true variance ( $\sigma^2$ ).

Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

# Estimators

- Because estimators are random variables, they have a distribution.
- Our knowledge about the underlying world that produced our data is governed by these distributions.

# Using the T-Distribution

Define

$$t = \frac{\bar{x} - \mu}{s/\sqrt{N}}$$
$$t \sim \text{Student's } t(N - 1)$$